LUND UNIVERSITY

Written Examination
Group and Ring Theory
MATP33
Wednesday, 25 May 2022
Duration: 08:00–13:00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

No books, notes, computational devices, etc. are allowed. Use clear handwriting and give clear careful motivations. Please write your anonymous identification number on each sheet of paper.

1. Find the number of different necklaces with p spherical beads, for p an odd prime, where the beads can have any of n different colours. Two necklaces are considered the same, if one necklace can be transformed into the other under rotations and reflections.

2. Let G be a group of order $351 = 3^3 \cdot 13$. Prove that G has a normal Sylow p-subgroup for some prime p dividing $|G|$.

3. Let $R = M_n(F)$ be the matrix ring over a field F, and for $1 \leq i, k \leq n$, denote by e_{ik} the matrix with 1 at the (i, k) entry and 0 elsewhere. For a fixed choice of i and k, is $M = Re_{ik}$ a simple R-module?

4. Let M be an Artinian R-module. Show that every quotient module of M is finitely cogenerated.

5. Determine the rational canonical form of

$$\begin{pmatrix} 2 & -2 & 14 \\ 0 & 3 & -7 \\ 0 & 0 & 2 \end{pmatrix}$$

considered as a complex matrix.

6. Let R be a Noetherian ring with unity. Prove that the polynomial ring $R[x]$ is also a Noetherian ring.